# GEOMETRIC REALIZATIONS OF ABSOLUTE GALOIS GROUPS IN PERFECTOID GEOMETRY

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Abstract. We generalize Weinstein's geometric realization of absolute Galois groups to a class of complete rank-one nonarchimedean fields of mixed characteristic. Let $E_\infty/E$ be a Galois algebraic extension whose completion $L = \widehat{E_\infty}$ is perfectoid, and let $C = \widehat{\overline{E}}$. Using the product of $\mathrm{Spa}(C^\flat)$ and $\mathrm{Spa}(L^\flat)$ in the category of perfectoid spaces, a partial-Frobenius quotient, and Galois descent, we construct a sheaf $Z_E$ on the pro-étale site over $C$ such that

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}(E).$$

Consequently,

$$\pi_1^{\text{ét}}(Z_E) \cong G_E$$

after choosing a geometric base point. We apply the construction to the two-dimensional local field $\mathbb{Q}_p\{\{X\}\}$ via its natural Kummer tower.

## 1. Introduction

Let $E/\mathbb{Q}_p$ be a finite extension. In [9], Weinstein constructed a geometric object $Z_E$, defined over a complete algebraically closed extension $C/E$, whose étale fundamental group recovers the absolute Galois group of $E$. More precisely, he proved an equivalence of Galois categories

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}(E).$$

The point of this construction is not merely to realize $G_E$ as a fundamental group: the Fargues–Fontaine curve already provides such a realization. Rather, Weinstein's construction produces an object defined over an algebraically closed nonarchimedean field, with the arithmetic information encoded in its geometry and group action.

The purpose of this paper is to extend this construction beyond ordinary local fields. Our main example is the two-dimensional local field

$$E = \mathbb{Q}_p\{\{X\}\}.$$

More generally, the same construction applies to a complete rank-one nonarchimedean field of mixed characteristic admitting a Galois algebraic extension whose completion is perfectoid; see Section 6 for the precise hypotheses.

**Theorem A.** *Let*

$$E = \mathbb{Q}_p\{\{X\}\}.$$

*There exists a geometric object $Z_E$, defined over the complete algebraically closed field $\widehat{\overline{E}}$, such that*

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}(E).$$

*In particular, after choosing a geometric base point,*

$$\pi_1^{\text{ét}}(Z_E) \cong G_E.$$

*More generally, the same conclusion holds for every complete rank-one nonarchimedean field $E$ of mixed characteristic satisfying the hypotheses of Theorem 7.1.*

The organization of the paper is the following: In Section 2, we recall the construction of the adic Fargues–Fontaine curve and the basic results which will be used later. Section 3 reviews Weinstein's realization of $G_E$ for an ordinary local field. In Section 4, we review Kedlaya's base change lemma, which forms the main geometric input of our construction, following from the universal geometric simple connectivity theorem. Section 5 contains the two arithmetic descent statements needed later: invariance of finite étale covers under completion and infinite Galois descent. Finally, in Section 6 we construct the space $Z_E$, prove the main theorem by combining these ingredients, and apply the general construction to $E = \mathbb{Q}_p\{\{X\}\}$.

## 2. Adic Spaces and Perfectoid Spaces

We recall the basic definitions of adic and perfectoid spaces, and some results on the existence of fiber products that will be used below. Our main references are [1] and [2].

2.1. **Basic notions and categories.** A *Huber ring* is a topological ring $A$ containing an open subring $A_0$ whose topology is $I$-adic for some finitely generated ideal $I \subseteq A_0$. Such an $A_0$ is called a *ring of definition*, and $I$ an *ideal of definition*. A Huber ring is called a *Tate ring* if it contains a topologically nilpotent unit; such an element is also called a *pseudo-uniformizer*.

Write $A^\circ$ for the subring of power-bounded elements of $A$. A *Huber pair* $(A, A^+)$ consists of a Huber ring $A$ and an open, integrally closed subring $A^+ \subseteq A^\circ$. Its *adic spectrum* $\mathrm{Spa}(A, A^+)$ has as points equivalence classes of continuous valuations on $A$ which are bounded by 1 on $A^+$. An *adic space* is a locally topologically ringed space equipped with valuations, locally isomorphic to the adic spectrum of a sheafy Huber pair. Morphisms are morphisms of locally topologically ringed spaces compatible with these valuations.

**Example (the generic fiber of a Tate Huber pair).** Let $(A, A^+)$ be a complete Tate Huber pair such that $\mathrm{Spa}(A, A^+)$ is an adic space, and choose a topologically nilpotent unit $\varpi \in A^+$. Give $A^+$ the topology induced from $A$. Since $A^+$ is open, we have $A = A^+[1/\varpi]$, and there is a natural identification

$$\mathrm{Spa}(A, A^+) \cong \mathrm{Spa}(A^+, A^+) \setminus V(\varpi)$$

A complete Tate ring $A$ is called *perfectoid* if it is uniform and admits a pseudo-uniformizer $\varpi$ such that $\varpi^p$ divides $p$ in $A^\circ$ and the Frobenius map

$$A^\circ/\varpi \longrightarrow A^\circ/\varpi^p, \qquad x \longmapsto x^p,$$

is an isomorphism. In characteristic $p$, a complete Tate ring is perfectoid if and only if it is a perfect ring. A *perfectoid space* is an adic space locally isomorphic to $\mathrm{Spa}(A, A^+)$ for an affinoid perfectoid Huber pair. We denote by **Adic** the category of adic spaces, by **Perfd** its full subcategory of perfectoid spaces, and by **Perf** the full subcategory of perfectoid spaces of characteristic $p$.

Perfectoid spaces provide a bridge between mixed-characteristic and characteristic-$p$ geometry. More precisely, tilting associates to each perfectoid space $X$ a characteristic-$p$ perfectoid space $X^\flat$, functorially in $X$. This is not simply a forgetful functor, it induces an equivalence of categories

$$\mathbf{Perfd}_{/X} \simeq \mathbf{Perf}_{/X^\flat}.$$

These equivalences, collectively referred to as *tilting equivalences*, allow many geometric constructions over $X$ to be studied over $X^\flat$. We refer to [12] for details.

2.2. **Fiber products.** Unlike in the category of schemes, fiber products of arbitrary adic spaces need not exist. The following gives a useful sufficient criterion.

**Proposition 2.1** (Huber [1], Proposition 3.7)**.** *Let $X_1, X_2, X_3$ be adic spaces with morphisms $f_i\colon X_i \to X_3$ for $i = 1, 2$. Suppose that $f_1$ is locally of finite type, that $f_2$ is adic, and that every point of $X_2$ has an open affinoid neighborhood $U$ for which $\mathcal{O}_{X_2}(U)$ has a Noetherian ring of definition or is a strongly Noetherian Tate ring. Then the fiber product*

$$X_1 \times_{X_3} X_2$$

*exists in* **Adic**. *Its projection to $X_2$ is locally of finite type and its projection to $X_1$ is adic.*

For instance, if $K$ is a complete nontrivially valued nonarchimedean field, the fiber product

$$\mathrm{Spa}(\mathbb{Z}[T_1, T_2, \ldots], \mathbb{Z}) \times_{\mathrm{Spa}(\mathbb{Z},\mathbb{Z})} \mathrm{Spa}(K, \mathcal{O}_K)$$

does not exist in **Adic**; see [3, Lecture 4, Section 4.1].

In contrast, fiber products exist in the category of perfectoid spaces.

**Proposition 2.2** (Fargues [2], Proposition 3.9.8)**.** *The category* **Perfd** *admits fiber products. For any morphisms of perfectoid spaces*

$$X_1 \longrightarrow S \longleftarrow X_2,$$

*the fiber product $X_1 \times_S X_2$ exists in* **Perfd**. *It is affinoid perfectoid if $X_1, X_2, S$ are affinoid perfectoid, and fiber products commute with tilting. In particular,* **Perf** *also admits fiber products.*

*Proof.* By the tilting equivalence, it suffices to prove the assertion in characteristic $p$. Indeed, tilting induces an equivalence between the categories of perfectoid spaces over $S$ and over $S^\flat$, and equivalences of categories preserve fiber products.

We first assume that all three spaces are affinoid. Write

$$X_i = \mathrm{Spa}(A_i, A_i^+), \quad i = 1, 2, \qquad S = \mathrm{Spa}(A_3, A_3^+).$$

Choose a topologically nilpotent unit $\varpi \in A_3^+$. Its images in $A_1$ and $A_2$ are again topologically nilpotent units.

Let

$$R_0 = \mathrm{im}\left(A_1^+ \otimes_{A_3^+} A_2^+ \longrightarrow A_1 \otimes_{A_3} A_2\right),$$

and let $R$ be its $\varpi$-adic completion. Put

$$A = R[1/\varpi],$$

endowed with the Tate topology for which $R$ is a ring of definition. Let $A^+$ be the closure in $A$ of the integral closure of the image of $R_0$. The completed tensor product construction gives natural morphisms

$$\mathrm{Spa}(A, A^+) \longrightarrow X_i, \qquad i = 1, 2,$$

compatible with the morphisms to $S$.

Since the rings $A_i$ are perfectoid of characteristic $p$, their Frobenius endomorphisms are bijective. The same is true of $A_i^+$, since these are integrally closed in $A_i^\circ$. Consequently, the Frobenius endomorphism of $R_0$ is bijective.

Moreover, Frobenius and its inverse are continuous for the $\varpi$-adic topology. Thus Frobenius extends to an automorphism of $R$, and hence of $A$. Therefore $A$ is a perfect complete Tate $\mathbb{F}_p$-algebra, and hence perfectoid. Since $A$ is uniform, the subring $A^+$ is open, bounded, and integrally closed in $A^\circ$. Hence $(A, A^+)$ is an affinoid perfectoid Huber pair.

We now verify the universal property of the fiber product. Let

$$T = \mathrm{Spa}(C, C^+)$$

be an affinoid perfectoid space equipped with compatible morphisms $f_i : T \to X_i$ over $S$. These correspond to continuous homomorphisms of Huber pairs

$$\phi_i : (A_i, A_i^+) \longrightarrow (C, C^+), \qquad i = 1, 2,$$

whose restrictions to $(A_3, A_3^+)$ coincide.

By the universal property of the tensor product, the maps $\phi_i$ induce a homomorphism

$$\phi_0 : R_0 \longrightarrow C^+, \qquad a_1 \otimes a_2 \longmapsto \phi_1(a_1)\phi_2(a_2).$$

Since $C^+$ is bounded and $\phi_0(R_0) \subseteq C^+$, while $\phi_0(\varpi)$ is topologically nilpotent, we have

$$\phi_0(\varpi)^n C^+ \longrightarrow 0$$

as $n \to \infty$. Consequently, $\phi_0$ is continuous for the $\varpi$-adic topology on $R_0$.

As $C^+$ is closed in the complete ring $C$, the homomorphism $\phi_0$ extends uniquely to a continuous homomorphism

$$\phi^+ : R = \widehat{R_0}^{(\varpi)} \longrightarrow C^+.$$

Moreover, $\phi^+(\varpi)$ is a unit in $C$. Hence $\phi^+$ extends uniquely to

$$\phi : A = R[1/\varpi] \longrightarrow C.$$

This homomorphism is continuous for the Tate topologies, since it maps the ring of definition $R$ into the bounded subring $C^+$ and sends the topologically nilpotent unit $\varpi$ to a topologically nilpotent unit of $C$.

We claim that $\phi(A^+) \subseteq C^+$. Indeed, by definition, $A^+$ is the closure of the integral closure of the image of $R_0$ in $A$. Since $\phi(R_0) \subseteq C^+$ and $C^+$ is integrally closed in $C$, every element integral over the image of $R_0$ is mapped into $C^+$. The claim follows because $C^+$ is closed and $\phi$ is continuous.

Thus $\phi$ defines a homomorphism of Huber pairs

$$(A, A^+) \longrightarrow (C, C^+),$$

and hence a morphism

$$f : T \longrightarrow \mathrm{Spa}(A, A^+)$$

whose compositions with the natural projections coincide with $f_1$ and $f_2$.

The morphism $f$ is unique. Indeed, any such morphism must induce the homomorphism $\phi_0$ on $R_0$, by the universal property of the tensor product. Since $R_0$ is dense in $R$ and $A = R[1/\varpi]$, its extension to $A$ is uniquely determined.

Therefore, for every affinoid perfectoid space $T$, we obtain a natural bijection

$$\mathrm{Hom}_{\mathbf{Perfd}}(T, \mathrm{Spa}(A, A^+)) \cong \mathrm{Hom}_{\mathbf{Perfd}}(T, X_1) \times_{\mathrm{Hom}_{\mathbf{Perfd}}(T,S)} \mathrm{Hom}_{\mathbf{Perfd}}(T, X_2).$$

By gluing on affinoid perfectoid coverings, the same universal property holds for arbitrary perfectoid spaces $T$. This proves

$$X_1 \times_S X_2 \cong \mathrm{Spa}(A, A^+)$$

in the category of perfectoid spaces.

The assertion for arbitrary perfectoid spaces $X_1, X_2, S$ follows by gluing the local constructions. Finally, the general characteristic case follows from the tilting equivalence, which also shows that fiber products commute with tilting. □

*Remark.* The fiber product in Proposition 2.2 is taken in **Perfd** and need not satisfy the universal property in **Adic**; see [2, Remark 3.9.9]. Moreover, **Perf** has no terminal object, so the existence of fiber products in that category does not by itself imply the existence of absolute products.

We next record an auxiliary result concerning the analytic locus of an adic spectrum in characteristic $p$.

**Proposition 2.3** (Fargues [2], Proposition 3.9.6)**.** *Let $R$ be a perfect $\mathbb{F}_p$-algebra, equipped with the $I$-adic topology for some finitely generated ideal $I \subseteq R$. Then its analytic locus*

$$\mathrm{Spa}(R,R)^a = \mathrm{Spa}(R,R) \setminus V(I)$$

*is a perfectoid space.*

*Proof.* Write $I = (f_1, \dots, f_r)$. Since $I$ is an ideal of definition, the analytic locus of $\mathrm{Spa}(R,R)$ is precisely

$$\mathrm{Spa}(R,R)^a = \mathrm{Spa}(R,R) \setminus V(I).$$

For each $i = 1, \dots, r$, consider the rational open subset

$$U_i = \{x \in \mathrm{Spa}(R,R) : |f_j(x)| \leq |f_i(x)| \neq 0 \text{ for all } j\}.$$

These rational subsets form an open covering of $\mathrm{Spa}(R,R)^a$.

Fix $i$. Put

$$B_i = R\left[\frac{f_1}{f_i}, \dots, \frac{f_r}{f_i}\right] \subseteq R[1/f_i],$$

and let $C_i$ be the completion of $R[1/f_i]$ for the Tate topology having $B_i$ as a ring of definition and $f_i$ as a topologically nilpotent unit. By the construction of rational localizations, $U_i$ is represented by a Huber pair $(C_i, C_i^+)$.

We claim that $C_i$ is perfect. Since $R$ is perfect, the Frobenius endomorphism

$$F : R[1/f_i] \longrightarrow R[1/f_i], \qquad a \longmapsto a^p$$

is bijective.

Clearly $F(B_i) \subseteq B_i$, so $F$ is continuous. On the other hand, its inverse satisfies

$$F^{-1}(B_i) \subseteq f_i^{-r} B_i.$$

Indeed, $F^{-1}(B_i)$ is generated over $B_i$ by the elements $(f_j/f_i)^{1/p}$, and is generated as a $B_i$-module by the monomials

$$\prod_{j=1}^{r} \left(\frac{f_j}{f_i}\right)^{e_j/p}, \qquad 0 \leq e_j < p.$$

Each such monomial belongs to $f_i^{-r}R$, because $R$ is perfect.

Consequently,

$$F^{-1}(f_i^{pN} B_i) \subseteq f_i^{N-r} B_i$$

for every $N \geq 0$. Thus $F^{-1}$ is also continuous.

It follows that Frobenius extends to a topological automorphism of $C_i$. Hence $C_i$ is a perfect complete Tate $\mathbb{F}_p$-algebra and hence perfectoid. Therefore each $U_i$ is an affinoid perfectoid space, and so $\mathrm{Spa}(R,R)^a$ is a perfectoid space. □

For characteristic-$p$ perfectoid spaces, the absolute product admits a stronger universal property.

**Proposition 2.4** (Fargues [2], Proposition 3.9.10)**.** *Let $k$ be a perfect field of characteristic $p$, equipped with the discrete topology, and let $X_1,\dots,X_n$ be $k$-perfectoid spaces. Then*

$$X := X_1 \times_{\mathrm{Spa}(k,k)} \cdots \times_{\mathrm{Spa}(k,k)} X_n$$

*exists in* **Adic** *and is a perfectoid space.*

*Suppose in addition that $X_i = \mathrm{Spa}(A_i, A_i^+)$ is affinoid perfectoid for every $i$. Choose topologically nilpotent units $\varpi_i \in A_i^+$, and put*

$$I = (\varpi_1,\dots,\varpi_n) \subseteq A_1^+ \otimes_k \cdots \otimes_k A_n^+, \qquad R = \left(A_1^+ \otimes_k \cdots \otimes_k A_n^+\right)^\wedge_I.$$

*Equipping $R$ with its $I$-adic topology, one has*

$$\boxed{X \cong \mathrm{Spa}(R,R) \setminus V(\varpi_1 \cdots \varpi_n).}$$

*Proof.* We first treat the case in which all $X_i$ are affinoid and use the notation of the proposition. The rings $A_i^+$ are perfect: indeed, if $a \in A_i^+$, its unique $p$-th root in the perfect ring $A_i$ is power-bounded and integral over $A_i^+$, hence lies in $A_i^+$. Consequently, $B := A_1^+ \otimes_k \cdots \otimes_k A_n^+$ is perfect, as $k$ is perfect. The Frobenius automorphism of $B$ and its inverse are continuous for the $I$-adic topology: the ideals

$$I = (\varpi_1,\dots,\varpi_n) \quad \text{and} \quad I^{1/p} = (\varpi_1^{1/p},\dots,\varpi_n^{1/p})$$

define the same topology. They therefore extend to inverse continuous endomorphisms of $R = \widehat{B}_I$, proving that $R$ is perfect.

By Proposition 2.3, $\mathrm{Spa}(R,R) \setminus V(I)$ is perfectoid. Since $V(I) \subseteq V(\varpi_1 \cdots \varpi_n)$, the open subspace

$$U := \mathrm{Spa}(R,R) \setminus V(\varpi_1 \cdots \varpi_n)$$

is perfectoid as well.

We verify the universal property in **Adic**. For every $i$, $A_i = A_i^+[1/\varpi_i]$, and the generic-fiber identification from the preceding subsection gives

$$X_i \cong \mathrm{Spa}(A_i^+, A_i^+) \setminus V(\varpi_i).$$

The canonical continuous maps $A_i^+ \to R$, together with the nonvanishing of the $\varpi_i$ on $U$, induce morphisms $U \to X_i$ over $\mathrm{Spa}(k,k)$.

Conversely, let $T$ be an adic space equipped with compatible morphisms $T \to X_i$ over $\mathrm{Spa}(k,k)$. Working locally on $T$, these give continuous maps from the Huber pairs $(A_i, A_i^+)$ to the local affinoid Huber pairs of $T$. Their restrictions to $A_i^+$ combine into a $k$-algebra map $B \to \mathcal{O}_T^+$, which is continuous for the $I$-adic topology: the images of $\varpi_i$ are topologically nilpotent, and the image of $B$ is bounded. By completion it extends uniquely to a continuous map $R \to \mathcal{O}_T^+$ locally on $T$. This gives a morphism $T \to \mathrm{Spa}(R,R)$, whose image lies in $U$ because every $\varpi_i$ is invertible on $T$. Uniqueness follows from the universal property of the tensor product and the density of $B$ in $R$. The local constructions are compatible with restrictions and thus glue, establishing the required universal property.

For arbitrary perfectoid spaces $X_i$, the construction is local on the factors and the resulting spaces glue along their open intersections. The assertion follows. □

*Remark.* The formula uses only the integral subrings $A_i^+$ and their topologies. Indeed, $A_i = A_i^+[1/\varpi_i]$ for every $i$, so the Tate rings $A_i$ can be recovered from these data.

We now consider a related special case in which one of the factors is not perfectoid. The following result will be used later in our study of Fargues–Fontaine curves.

**Proposition 2.5.** *Let $k = \mathbb{F}_q$ and $E = k((\pi))$. Let $F$ be a perfectoid field of characteristic $p$ containing $k$, and choose a topologically nilpotent nonzero element $\varpi_F \in \mathcal{O}_F$. Put*

$$R = \mathcal{O}_F \hat{\otimes}_k \mathcal{O}_E \cong \mathcal{O}_F[[\pi]],$$

*be the completed tensor product for the $(\varpi_F, \pi)$-adic topology. Then the fiber product*

$$\mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(k,k)} \mathrm{Spa}(E, \mathcal{O}_E)$$

*exists in* **Adic**, *and there is a canonical isomorphism*

$$\mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(k,k)} \mathrm{Spa}(E, \mathcal{O}_E) \cong \mathrm{Spa}(R, R) \setminus V(\varpi_F \pi).$$

*Proof.* Put

$$U = \mathrm{Spa}(R, R) \setminus V(\varpi_F \pi).$$

Since $R \cong \mathcal{O}_F[[\pi]]$, the space $U$ is the punctured open unit disc over $F$. More precisely, it is covered by the affinoid annuli

$$U_{m,n} = \{x \in U : |\varpi_F(x)|^m \le |\pi(x)|, \quad |\pi(x)|^n \le |\varpi_F(x)|\}, \qquad m, n \ge 1.$$

In particular, $U$ is an adic space.

The canonical homomorphisms $\mathcal{O}_F \to R$ and $\mathcal{O}_E \to R$ induce morphisms

$$p_F : U \longrightarrow \mathrm{Spa}(F, \mathcal{O}_F), \qquad p_E : U \longrightarrow \mathrm{Spa}(E, \mathcal{O}_E),$$

since $\varpi_F$ and $\pi$ are invertible on $U$. These morphisms agree over $\mathrm{Spa}(k, k)$.

We verify the universal property in **Adic**. Let $T$ be an adic space equipped with compatible morphisms

$$f_F : T \longrightarrow \mathrm{Spa}(F, \mathcal{O}_F), \qquad f_E : T \longrightarrow \mathrm{Spa}(E, \mathcal{O}_E).$$

Since the assertion is local on $T$, we may assume that $T = \mathrm{Spa}(C, C^+)$ is affinoid, with $C$ complete.

The morphisms $f_F$ and $f_E$ correspond to continuous homomorphisms of Huber pairs

$$(F, \mathcal{O}_F) \longrightarrow (C, C^+), \qquad (E, \mathcal{O}_E) \longrightarrow (C, C^+).$$

Their restrictions to $k$ coincide. Hence the universal property of the tensor product gives a homomorphism

$$\phi_0 : \mathcal{O}_F \otimes_k \mathcal{O}_E \longrightarrow C^+, \qquad a \otimes b \longmapsto f_F^\sharp(a) f_E^\sharp(b).$$

We claim that $\phi_0$ is continuous for the $(\varpi_F, \pi)$-adic topology. Indeed, the images of $\mathcal{O}_F$ and $\mathcal{O}_E$ are bounded in $C$, so the image of their tensor product is bounded. Moreover, the images of $\varpi_F$ and $\pi$ are topologically nilpotent. Consequently,

$$\phi_0\big((\varpi_F, \pi)^n\big) \longrightarrow 0$$

uniformly on the bounded image of the tensor product as $n \to \infty$.

Since $C$ is complete, $\phi_0$ extends uniquely to a continuous homomorphism

$$\phi : R \longrightarrow C.$$

As $C^+$ is an open additive subgroup of $C$, it is closed. Thus $\phi(R) \subseteq C^+$.

The homomorphism $\phi$ induces a morphism

$$f : T \longrightarrow \mathrm{Spa}(R, R).$$

Since the images of $\varpi_F$ and $\pi$ are units in $C$, this morphism factors uniquely through $U$.

By construction, we have

$$p_F \circ f = f_F, \qquad p_E \circ f = f_E.$$

Conversely, every morphism $T \to U$ induces such a compatible pair. The construction is unique by the universal properties of the tensor product and completion, together with the density of $\mathcal{O}_F \otimes_k \mathcal{O}_E$ in $R$.

Thus $U$ satisfies the required universal property for affinoid test spaces. The general case follows by gluing over affinoid coverings of $T$. Hence

$$U \cong \mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(k,k)} \mathrm{Spa}(E, \mathcal{O}_E)$$

in **Adic**. □

**Proposition 2.6.** *Let $k = \mathbb{F}_q$, $E = k((\pi))$, and let $F$ be a complete nonarchimedean field of characteristic $p$ containing $k$. Choose a topologically nilpotent nonzero element $\varpi_F \in \mathcal{O}_F$. Let $A$ be an $E$-affinoid algebra, and choose a complete bounded ring of definition $A_0 \subseteq A$. Put*

$$B_0 = \mathcal{O}_F \widehat{\otimes}_k A_0.$$

*Then the fiber product*

$$\mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(k,k)} \mathrm{Spa}(A, A^\circ)$$

*exists in* **Adic**, *and there is a canonical isomorphism*

$$\begin{aligned} &\mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(k,k)} \mathrm{Spa}(A, A^\circ) \\ &\quad \cong \mathrm{Spa}(B_0, B_0) \setminus V(\varpi_F \pi), \end{aligned}$$

*where the right-hand side is endowed with the adic-space structure given by its rational affinoid charts.*

*In particular, if $A$ is uniform, one may take $A_0 = A^\circ$, obtaining*

$$\begin{aligned} &\mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(k,k)} \mathrm{Spa}(A, A^\circ) \\ &\quad \cong \mathrm{Spa}\left(\mathcal{O}_F \widehat{\otimes}_k A^\circ, \mathcal{O}_F \widehat{\otimes}_k A^\circ\right) \setminus V(\varpi_F \pi). \end{aligned}$$

*Proof.* Since $A$ is an $E$-affinoid algebra, we may choose $A_0$ to be a Noetherian ring of definition. We first observe that $A^\circ$ is the integral closure of $A_0$ in $A$.

Indeed, if $a \in A^\circ$, then the powers of $a$ form a bounded subset of $A$. Hence there exists $N \geq 0$ such that

$$a^n \in \pi^{-N} A_0 \qquad (n \geq 0).$$

Since $A_0$ is Noetherian, the $A_0$-module generated by all powers of $a$ is finite. Thus $a$ is integral over $A_0$. Conversely, every element integral over $A_0$ is power-bounded.

Put

$$U = \mathrm{Spa}(B_0, B_0) \setminus V(\varpi_F \pi).$$

We first verify that $U$ is an adic space. For positive integers $m, n$, let

$$U_{m,n} = \left\{ x \in U : |\varpi_F(x)|^m \leq |\pi(x)|, \quad |\pi(x)|^n \leq |\varpi_F(x)| \right\}.$$

Since both $\varpi_F$ and $\pi$ are topologically nilpotent in $B_0$, the continuity of valuations implies that the $U_{m,n}$ cover $U$.

Moreover, these are rational subsets. Let $D_{m,n}$ denote the $F$-affinoid algebra of the closed annulus

$$|\varpi_F|^m \leq |t|, \qquad |t|^n \leq |\varpi_F|,$$

viewed as an $E$-algebra via $\pi \mapsto t$. The underlying Tate ring of the rational localization corresponding to $U_{m,n}$ is naturally

$$D_{m,n}\widehat{\otimes}_E A.$$

Since $A$ is an $E$-affinoid algebra, this completed tensor product is an $F$-affinoid algebra. In particular, it is strongly Noetherian and sheafy. Consequently, the rational localizations above endow $U$ with the structure of an adic space.

The canonical homomorphisms $\mathcal{O}_F \to B_0$ and $A_0 \to B_0$ induce a morphism

$$p_F : U \longrightarrow \mathrm{Spa}(F, \mathcal{O}_F),$$

and a morphism

$$p_A : U \longrightarrow \mathrm{Spa}(A, A^\circ).$$

For the latter, we use $A = A_0[1/\pi]$ and the fact that $A^\circ$ is integral over $A_0$.

It remains to verify the universal property. Let

$$T = \mathrm{Spa}(C, C^+)$$

be an affinoid adic space with compatible morphisms

$$f_F : T \to \mathrm{Spa}(F, \mathcal{O}_F), \qquad f_A : T \to \mathrm{Spa}(A, A^\circ)$$

over $\mathrm{Spa}(k, k)$.

These induce continuous homomorphisms

$$\mathcal{O}_F \longrightarrow C^+, \qquad A_0 \longrightarrow C^+.$$

Their images are bounded in $C$, since $\mathcal{O}_F$ and $A_0$ are bounded in their respective Tate rings.

By the universal property of the tensor product, we obtain a homomorphism

$$\phi_0 : \mathcal{O}_F \otimes_k A_0 \longrightarrow C^+.$$

The image of $\phi_0$ is bounded. Moreover, the images of $\varpi_F$ and $\pi$ are topologically nilpotent units in $C$.

It follows that $\phi_0$ is continuous for the $(\varpi_F, \pi)$-adic topology. Since $C$ is complete, it extends uniquely to a continuous homomorphism

$$\phi : B_0 \longrightarrow C.$$

As $C^+$ is closed in $C$, we have $\phi(B_0) \subseteq C^+$.

The homomorphism $\phi$ therefore induces a morphism

$$f : T \longrightarrow \mathrm{Spa}(B_0, B_0).$$

Since $\varpi_F$ and $\pi$ map to units, the morphism $f$ factors through $U$.

By construction, the compositions of $f$ with $p_F$ and $p_A$ are precisely $f_F$ and $f_A$.

Conversely, any morphism $T \to U$ induces such a compatible pair of morphisms. The uniqueness follows from the universal properties of tensor products and completions, since $\mathcal{O}_F \otimes_k A_0$ is dense in $B_0$.

Thus $U$ satisfies the required universal property for affinoid test spaces. The general case follows by gluing. □

## 3. Fargues–Fontaine curves

Let $E$ be a local field of residue characteristic $p$, with ring of integers $\mathcal{O}_E$, residue field $k_E \cong \mathbb{F}_q$, and a fixed uniformizer $\pi$. Let $F$ be an algebraically closed nonarchimedean field of characteristic $p$ equipped with an embedding $k_E \hookrightarrow F$. We fix a pseudo-uniformizer $\varpi_F \in \mathcal{O}_F$. For $a \in \mathbb{Q}_{>0}$, choose $\varpi_F^a \in F$ such that $|\varpi_F^a| = |\varpi_F|^a$.

We briefly recall the construction of the adic Fargues–Fontaine curve associated to the pair $(F, E)$.

3.1. **The mixed-characteristic case.** Suppose first that $\mathrm{char}(E) = 0$. Put

$$A_{\mathrm{inf},F,E} := W_{\mathcal{O}_E}(\mathcal{O}_F) := W(\mathcal{O}_F) \otimes_{W(k_E)} \mathcal{O}_E.$$

Since $F$ is perfect, every element of $A_{\mathrm{inf},F,E}$ admits a unique Teichmüller expansion

$$x = \sum_{n \geq 0} \pi^n [x_n], \qquad x_n \in \mathcal{O}_F.$$

We equip $A_{\mathrm{inf},F,E}$ with the $(\pi, [\varpi_F])$-adic topology and define

$$\mathcal{Y}_{F,E} := \mathrm{Spa}(A_{\mathrm{inf},F,E}, A_{\mathrm{inf},F,E}) \setminus V\big(\pi[\varpi_F]\big).$$

For a closed interval $I = [a, b] \subset (0, \infty)$ with $a, b \in \mathbb{Q}$, define

$$\mathcal{Y}_{F,E}^I := U\left(\frac{\{\pi, [\varpi_F^a]\}}{[\varpi_F^a]}, \frac{\{\pi, [\varpi_F^b]\}}{\pi}\right) = \left\{x \in \mathcal{Y}_{F,E} \,\middle|\, |[\varpi_F^b](x)| \leq |\pi(x)| \leq |[\varpi_F^a](x)|\right\}.$$

As $I$ varies, these rational subsets cover $\mathcal{Y}_{F,E}$.

Let $R_{F,E}^I$ be the $\pi$-adic completion of

$$A_{\mathrm{inf},F,E}\left[\frac{\pi}{[\varpi_F^a]}, \frac{[\varpi_F^b]}{\pi}\right],$$

and put

$$B_{F,E}^I := R_{F,E}^I[1/\pi],$$

with $B_{F,E}^{I,+}$ the integral closure of $R_{F,E}^I$ in $B_{F,E}^I$. Then

$$\mathcal{Y}_{F,E}^I = \mathrm{Spa}\left(B_{F,E}^I, B_{F,E}^{I,+}\right).$$

The $q$-power Frobenius of $F$ induces an automorphism

$$\varphi : A_{\mathrm{inf},F,E} \longrightarrow A_{\mathrm{inf},F,E}, \qquad \sum_{n \geq 0} \pi^n [x_n] \longmapsto \sum_{n \geq 0} \pi^n [x_n^q],$$

and hence an automorphism of $\mathcal{Y}_{F,E}$.

3.2. **The equal-characteristic case.** Suppose now that $\mathrm{char}(E) = p$. After choosing $\pi$, we identify

$$E \simeq k_E((\pi)), \qquad \mathcal{O}_E \simeq k_E[[\pi]].$$

In this case put

$$A_{\mathrm{inf},F,E} := \mathcal{O}_F[[\pi]],$$

equipped with the $(\varpi_F, \pi)$-adic topology, and define

$$\mathcal{Y}_{F,E} := \mathrm{Spa}\big(\mathcal{O}_F[[\pi]], \mathcal{O}_F[[\pi]]\big) \setminus V(\pi\varpi_F).$$

For $I = [a, b] \subset (0, \infty)$ with $a, b \in \mathbb{Q}$, define

$$\mathcal{Y}^I_{F,E} := U\left(\frac{\{\pi, \varpi_F^a\}}{\varpi_F^a}, \frac{\{\pi, \varpi_F^b\}}{\pi}\right) = \left\{x \in \mathcal{Y}_{F,E} \,\middle|\, |\varpi_F^b(x)| \leq |\pi(x)| \leq |\varpi_F^a(x)|\right\}.$$

Again these rational subsets cover $\mathcal{Y}_{F,E}$. Let $R^I_{F,E}$ be the $\pi$-adic completion of

$$\mathcal{O}_F[[\pi]]\left[\frac{\pi}{\varpi_F^a}, \frac{\varpi_F^b}{\pi}\right],$$

put $B^I_{F,E} := R^I_{F,E}[1/\pi]$, and let $B^{I,+}_{F,E}$ be the integral closure of $R^I_{F,E}$ in $B^I_{F,E}$. Then

$$\mathcal{Y}^I_{F,E} = \mathrm{Spa}\left(B^I_{F,E}, B^{I,+}_{F,E}\right).$$

The $q$-power Frobenius on the coefficients induces

$$\varphi : \mathcal{O}_F[[\pi]] \longrightarrow \mathcal{O}_F[[\pi]], \qquad \sum_{n\geq 0} x_n \pi^n \longmapsto \sum_{n\geq 0} x_n^q \pi^n,$$

and hence an automorphism of $\mathcal{Y}_{F,E}$.

3.3. **The annular covering.** The rings $B^I_{F,E}$ admit an equivalent description as extended Robba rings. Put

$$B^b_{F,E} = \begin{cases} A_{\mathrm{inf},F,E}\left[\dfrac{1}{\pi}, \dfrac{1}{[\varpi_F]}\right], & \mathrm{char}(E) = 0, \\ A_{\mathrm{inf},F,E}\left[\dfrac{1}{\pi}, \dfrac{1}{\varpi_F}\right], & \mathrm{char}(E) = p. \end{cases}$$

Fix a normalization $0 < |\pi|_E < 1$. For $t > 0$, define

$$\lambda_t\left(\sum_{n \gg -\infty} \pi^n [x_n]\right) = \sup_n |\pi|_E^n |x_n|^t$$

in mixed characteristic, and

$$\lambda_t\left(\sum_{n \gg -\infty} x_n \pi^n\right) = \sup_n |\pi|_E^n |x_n|^t$$

in equal characteristic. For $I = [a, b]$, the ring $B^I_{F,E}$ is the completion with respect to the norms $\lambda_t$ for $t \in I$. Since $t \mapsto \log \lambda_t(x)$ is convex, it is equivalently the completion for the Banach norm

$$\lambda_I(x) := \max\{\lambda_a(x), \lambda_b(x)\}.$$

The rings $B^I_{F,E}$ are strongly noetherian. In particular,

$$\mathcal{Y}_{F,E} = \varinjlim_I \mathcal{Y}^I_{F,E} = \varinjlim_I \mathrm{Spa}\left(B^I_{F,E}, B^{I,+}_{F,E}\right)$$

is an adic space. The Frobenius satisfies

$$\varphi : B^I_{F,E} \longrightarrow B^{I/q}_{F,E}$$

and induces an automorphism of $\mathcal{Y}_{F,E}$.

The action of $\varphi^{\mathbb{Z}}$ on $\mathcal{Y}_{F,E}$ is properly discontinuous. We define the adic Fargues–Fontaine curve by

$$\mathcal{X}_{F,E} := \mathcal{Y}_{F,E}/\varphi^{\mathbb{Z}}.$$

The quotient $\mathcal{X}_{F,E}$ is a well-defined adic space.

3.4. **Results on the Fargues–Fontaine curve.** We recall some classical results on the Fargues–Fontaine curve. For $\lambda \in \mathbb{Q}$, let $\mathcal{O}(\lambda)$ denote the stable vector bundle of slope $\lambda$ on $\mathcal{X}_{F,E}$. If $\lambda = d/h$ with $(d,h) = 1$ and $h > 0$, then $\mathcal{O}(\lambda)$ has rank $h$ and degree $d$.

The classification of vector bundles in mixed characteristic is due to Fargues [4, Théorème 3.1]; the equal-characteristic analogue is due to Hartl–Pink [6, Theorem 11.1]. We use the uniform formulation of Fargues–Fontaine [5, Théorème 8.2.10]; see also [8, Theorem 4.6].

**Theorem 3.1.** *Every vector bundle $\mathcal{E}$ on $\mathcal{X}_{F,E}$ admits a decomposition*

$$\mathcal{E} \cong \bigoplus_{i=1}^{n} \mathcal{O}(\lambda_i), \qquad \lambda_i \in \mathbb{Q},$$

*where the multiset of slopes $\{\lambda_1, \dots, \lambda_n\}$ is uniquely determined by $\mathcal{E}$.*

We will also use the following computation of global sections. In mixed characteristic this is [9, Proposition 3.3.3]; the same statement in both characteristics is contained in [8, Lemma 4.5].

**Proposition 3.2.** *For $\lambda \in \mathbb{Q}$,*

$$H^0\big(\mathcal{X}_{F,E}, \mathcal{O}(\lambda)\big) \neq 0 \quad \Longleftrightarrow \quad \lambda \geq 0.$$

*Moreover,*

$$H^0\big(\mathcal{X}_{F,E}, \mathcal{O}_{\mathcal{X}_{F,E}}\big) = E.$$

Combining these results with the trace pairing gives the geometric simple connectivity of the Fargues–Fontaine curve. In mixed characteristic this is [9, Theorem 3.4.3]; the same argument applies in equal characteristic using the classification above.

**Theorem 3.3.** *Base extension induces an equivalence*

$$\mathbf{F\acute{E}t}(E) \xrightarrow{\sim} \mathbf{F\acute{E}t}\big(\mathcal{X}_{F,E}\big).$$

*In particular, after choosing a geometric point,*

$$\pi_1^{\text{ét}}\big(\mathcal{X}_{F,E}\big) \cong G_E.$$

In addition, we record the following identification, which will be used in Section 4.

**Proposition 3.4.** *Suppose that* $\mathrm{char}(E) = p$*, so that $E = \mathbb{F}_q((\pi))$, and let $A$ be an $E$-affinoid algebra. Then there is a canonical identification*

$$\begin{aligned}\mathcal{Y}_{F,E} \times_{\mathrm{Spa}(E,\mathcal{O}_E)} \mathrm{Spa}(A, A^\circ) &\cong \mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(\mathbb{F}_q,\mathbb{F}_q)} \mathrm{Spa}(A, A^\circ) \\ &\cong \mathrm{Spa}\big(\mathcal{O}_F \widehat{\otimes}_{\mathbb{F}_q} A^\circ, \mathcal{O}_F \widehat{\otimes}_{\mathbb{F}_q} A^\circ\big) \setminus V(\varpi_F \pi).\end{aligned}$$

*Proof.* Firstly, when $A = E$, Proposition 2.5 gives

$$\begin{aligned}&\mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(\mathbb{F}_q,\mathbb{F}_q)} \mathrm{Spa}(E, \mathcal{O}_E) \\ &\quad\cong \mathrm{Spa}\big(\mathcal{O}_F \widehat{\otimes}_{\mathbb{F}_q} \mathcal{O}_E, \mathcal{O}_F \widehat{\otimes}_{\mathbb{F}_q} \mathcal{O}_E\big) \setminus V(\varpi_F \pi) \\ &\quad\cong \mathcal{Y}_{F,E}.\end{aligned}$$

For the general case, the preceding identification and the associativity of fiber products yield

$$\begin{aligned}
&\mathcal{Y}_{F,E} \times_{\mathrm{Spa}(E,\mathcal{O}_E)} \mathrm{Spa}(A, A^\circ) \\
&\quad \cong \big(\mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(\mathbb{F}_q,\mathbb{F}_q)} \mathrm{Spa}(E, \mathcal{O}_E)\big) \times_{\mathrm{Spa}(E,\mathcal{O}_E)} \mathrm{Spa}(A, A^\circ) \\
&\quad \cong \mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathrm{Spa}(\mathbb{F}_q,\mathbb{F}_q)} \mathrm{Spa}(A, A^\circ) \\
&\quad \cong \mathrm{Spa}\big(\mathcal{O}_F \widehat{\otimes}_{\mathbb{F}_q} A^\circ, \mathcal{O}_F \widehat{\otimes}_{\mathbb{F}_q} A^\circ\big) \setminus V(\varpi_F \pi).
\end{aligned}$$

□

## 4. Realizing $G_E$ as a fundamental group

We review the construction of Weinstein [9, §3.5]. Throughout this section, let $E/\mathbb{Q}_p$ be a finite extension with residue field $k_E = \mathbb{F}_q$ and uniformizer $\pi$, and let $C$ be a complete algebraically closed extension of $E$.

4.1. **The Lubin–Tate universal cover.** Choose a power series

$$f(T) \in T\mathcal{O}_E[[T]], \qquad f(T) \equiv T^q \pmod{\pi},$$

and let $H_E = \mathrm{Spf}\, \mathcal{O}_E[[T]]$ be the Lubin–Tate formal $\mathcal{O}_E$-module associated to $f$, so that

$$[\pi]_{H_E}(T) = f(T).$$

Choose a compatible sequence of Lubin–Tate torsion points $t_1, t_2, \ldots$ satisfying

$$f(t_1) = 0, \qquad f(t_{n+1}) = t_n,$$

and put

$$E_n := E(t_n), \qquad E_\infty := \bigcup_{n \geq 1} E_n.$$

The completion $\widehat{E}_\infty$ is a perfectoid field.

Let

$$\widetilde{H}_E := \varprojlim_{[\pi]} H_E$$

be the universal cover of $H_E$. This is a formal $E$-vector space. Put

$$H_{E,0} := H_E \otimes_{\mathcal{O}_E} \mathbb{F}_q, \qquad \widetilde{H}_{E,0} := \widetilde{H}_E \otimes_{\mathcal{O}_E} \mathbb{F}_q.$$

Since $[\pi]_{H_{E,0}}(T) = T^q$, one has

$$\widetilde{H}_{E,0} \cong \mathrm{Spf}\, \mathbb{F}_q[[T^{1/q^\infty}]].$$

Moreover, Weinstein constructs an $E^\times$-equivariant isomorphism

$$\widetilde{H}_{E,0} \cong \mathrm{Spf}\, \mathcal{O}_{\widehat{E}_\infty^\flat},$$

where $\mathcal{O}_E^\times$ acts on $\widehat{E}_\infty$ through local class field theory and $\pi$ acts on $\widehat{E}_\infty^\flat$ by the $q$-power Frobenius [9, Proposition 3.5.3].

Let

$$\widetilde{H}_{E,C}^{\mathrm{ad}} := \Big(\widetilde{H}_E \widehat{\otimes}_{\mathcal{O}_E} \mathcal{O}_C\Big)_\eta^{\mathrm{ad}}, \qquad \widetilde{H}_{E,C}^{\mathrm{ad},*} := \widetilde{H}_{E,C}^{\mathrm{ad}} \setminus \{0\}.$$

Let $\varpi_C^\flat \in \mathcal{O}_{C^\flat}$ be a pseudo-uniformizer, and let $t \in \mathcal{O}_{\widehat{E}_\infty^\flat}$ be the pseudo-uniformizer arising from the compatible Lubin–Tate tower. By [9, Proposition 3.5.4], there are $E^\times$-equivariant isomorphisms

$$\left(\widetilde{H}_{E,C}^{\mathrm{ad},*}\right)^\flat \cong \left(\mathcal{Y}_{C^\flat,E}\widehat{\otimes}_E\widehat{E}_\infty\right)^\flat \cong \mathrm{Spa}(R,R^+),$$

where

$$R^+ := \mathcal{O}_{C^\flat}\widehat{\otimes}_{\mathbb{F}_q}\mathcal{O}_{\widehat{E}_\infty^\flat}, \qquad R := R^+\left[\frac{1}{\varpi_C^\flat \otimes t}\right].$$

Equivalently,

$$\mathrm{Spa}(R,R^+) = \mathrm{Spa}(R^+,R^+)\setminus V(\varpi_C^\flat \otimes t).$$

In fact, this gives a product interpretation of the tilted Lubin–Tate space: We write

$$\mathrm{Spa}(C^\flat,\mathcal{O}_{C^\flat}) \times_{\mathbb{F}_q} \mathrm{Spa}(\widehat{E}_\infty^\flat,\mathcal{O}_{\widehat{E}_\infty^\flat})$$

for the perfectoid space obtained from the completed tensor product over $\mathbb{F}_q$ as above. Then by an argument similar to the proof of Lemma 5.2(a) in the following section, we have

$$\left(\widetilde{H}_{E,C}^{\mathrm{ad},*}\right)^\flat \cong \mathrm{Spa}(C^\flat,\mathcal{O}_{C^\flat}) \times_{\mathbb{F}_q} \mathrm{Spa}(\widehat{E}_\infty^\flat,\mathcal{O}_{\widehat{E}_\infty^\flat}).$$

When $q = p$, this is precisely the absolute product of perfectoid spaces. For general $q = p^f$, the absolute product is constructed using the completed tensor product over $\mathbb{F}_p$, and the above $\mathbb{F}_q$-linear product is the clopen component determined by the chosen $\mathbb{F}_q$-structures.

4.2. **The space $Z_E$ and Weinstein's theorem.** Following Weinstein, we define

$$Z_E := \widetilde{H}_{E,C}^{\mathrm{ad},*}/E^\times$$

as a sheaf on the pro-étale site of perfectoid spaces over $C$. Writing

$$E^\times = \mathcal{O}_E^\times \times \pi^{\mathbb{Z}},$$

the action of $\pi^{\mathbb{Z}}$ is totally discontinuous, while

$$\widetilde{H}_{E,C}^{\mathrm{ad},*}/\pi^{\mathbb{Z}} \longrightarrow Z_E$$

is a pro-étale $\mathcal{O}_E^\times$-torsor.

For a group $G$ acting on a space $X$, write $\mathbf{F\acute{E}t}_G(X)$ for the category of $G$-equivariant finite étale covers of $X$.

**Theorem 4.1** (Weinstein)**.** *There is a natural equivalence*

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}(E).$$

*In particular,*

$$\pi_1^{\text{ét}}(Z_E) \cong G_E.$$

*Proof.* The result follows from the following chain of equivalences:

$$\begin{aligned}
\mathbf{F\acute{E}t}(Z_E) &\simeq \mathbf{F\acute{E}t}_{E^\times}\left(\widetilde{H}^{\mathrm{ad},*}_{E,C}\right)\\
&\simeq \mathbf{F\acute{E}t}_{E^\times}\left(\left(\widetilde{H}^{\mathrm{ad},*}_{E,C}\right)^\flat\right)\\
&\simeq \mathbf{F\acute{E}t}_{\mathcal{O}_E^\times\times(\varphi^{-1}\circ\varphi_q)^{\mathbb{Z}}}\left(\mathcal{Y}_{C^\flat,E}\widehat{\otimes}_E\widehat{E}_\infty\right)\\
&\simeq \mathbf{F\acute{E}t}_{\mathcal{O}_E^\times\times\varphi^{\mathbb{Z}}}\left(\mathcal{Y}_{C^\flat,E}\widehat{\otimes}_E\widehat{E}_\infty\right)\\
&\simeq \mathbf{F\acute{E}t}_{\mathcal{O}_E^\times}\left(\mathcal{X}_{C^\flat,E}\widehat{\otimes}_E\widehat{E}_\infty\right)\\
&\simeq \mathbf{F\acute{E}t}_{\mathcal{O}_E^\times}(\widehat{E}_\infty)\\
&\simeq \mathbf{F\acute{E}t}(E).
\end{aligned}$$

Since $\widetilde{H}^{\mathrm{ad},*}_{E,C}/\pi^{\mathbb{Z}} \to Z_E$ is a pro-étale $\mathcal{O}_E^\times$-torsor, finite étale covers of $Z_E$ is equivalent to $\mathcal{O}_E^\times$-equivariant finite étale covers of $\widetilde{H}^{\mathrm{ad},*}_{E,C}/\pi^{\mathbb{Z}}$. Since the action of $\pi$ on $\widetilde{H}^{\mathrm{ad},*}_{E,C}$ is discrete, this is again equivalent to $E^\times$-equivariant finite étale covers of $\widetilde{H}^{\mathrm{ad},*}_{E,C}$, and the first equivalence above holds. The second equivalence is the tilting equivalence, the third follows from the identification of $\left(\widetilde{H}^{\mathrm{ad},*}_{E,C}\right)^\flat$ in the end of the previous subsection. For the fourth equivalence, recall that the absolute $q$-power Frobenius $\varphi_q$ is an automorphism of every perfectoid space of characteristic $p$, and also of every finite étale cover of such a space. Hence, for any $\mathbb{F}_q$-linear automorphism $f$ of a perfectoid space, composing an $f$-equivariant structure on a finite étale cover with $\varphi_q$ gives an $(f\circ\varphi_q)$-equivariant structure, and this operation is invertible. Applying this with $f=\varphi^{-1}$ identifies $(\varphi^{-1}\circ\varphi_q)^{\mathbb{Z}}$-equivariant covers with $\varphi^{-1,\mathbb{Z}}$-equivariant covers, or equivalently with $\varphi^{\mathbb{Z}}$-equivariant covers. The fifth is the definition of the adic Fargues–Fontaine curve, while the last two equivalences follow from the geometric simple connectivity of $\mathcal{X}_{C^\flat,E}$ and Galois descent along the Lubin–Tate extension. □

## 5. Kedlaya's base change lemma

5.1. **Universal geometric simple connectivity.** We first recall the strengthened geometric simple connectivity theorem of Kedlaya [8, Theorem 9.1]. Let $F$ be an algebraically closed nonarchimedean field of characteristic $p$, let $E$ be a local field with residue field $\mathbb{F}_q$, and let $L$ be a complete algebraically closed overfield of $E$.

**Theorem 5.1** (Kedlaya)**.** *Every finite étale cover of*

$$\mathcal{X}_{F,E}\times_E L$$

*splits completely.*

*Proof.* We briefly recall the argument. By a tilting reduction, one may reduce to the case $\mathrm{char}(E)=0$. When $L$ is a completed algebraic closure of $E$, the assertion follows from the classification of vector bundles on the Fargues–Fontaine curve, as recalled in Section 2.

The general case is obtained by transfinite induction. It is enough to show that, if the assertion holds for $L$, then it continues to hold after replacing $L$ by the completed algebraic closure $L'$ of the completion of $L(T)$ with respect to an arbitrary multiplicative norm. Such a norm determines a point $x$ of type 2, 3, or 4, in the sense of Berkovich, on the adic affine

line over $L$. The point $x$ admits a fundamental system of étale neighborhoods $D$ which are smooth one-dimensional affinoid spaces over $L$.

Let

$$U' \longrightarrow \mathcal{X}_{F,E} \times_E L'$$

be a finite étale cover. This cover spreads out to a finite étale cover

$$U_D \longrightarrow \mathcal{X}_{F,E} \times_E D$$

for a sufficiently small étale neighborhood $D$ of $x$. The essential input is [8, Lemma 8.4]: after replacing $D$ by a further étale neighborhood $D' \to D$, the cover $U_D$ becomes split. This lemma is obtained from the analysis of the subsidiary radii of the corresponding $p$-adic differential equations. After base change to $L'$, the original cover therefore splits. The transfinite induction proves the theorem. □

5.2. **Products of perfectoid spaces.** Let **Pfd** denote the category of perfectoid spaces of characteristic $p$. We recall the construction of absolute products in **Pfd** and the associated partial Frobenius quotients [8, Lemma 9.3].

**Lemma 5.2** (Kedlaya)**.** *Let $X_1, \dots, X_n \in \mathbf{Pfd}$.*

(a) *The categorical product*

$$X := X_1 \times \cdots \times X_n$$

*exists in* **Pfd**. *More precisely, if*

$$X_i = \mathrm{Spa}(A_i, A_i^+)$$

*is affinoid perfectoid and $\varpi_i \in A_i$ is a topologically nilpotent unit, put*

$$A^+ := \widehat{A_1^+ \otimes_{\mathbb{F}_p} \cdots \otimes_{\mathbb{F}_p} A_n^+},$$

*where the completion is taken for the $(\varpi_1, \dots, \varpi_n)$-adic topology, and*

$$A := A^+[\varpi_1^{-1}, \dots, \varpi_n^{-1}].$$

*Then*

$$X_1 \times \cdots \times X_n \cong \left\{ x \in \mathrm{Spa}(A, A^+) \,\middle|\, |\varpi_i(x)| < 1 \text{ for } i = 1, \dots, n \right\}.$$

*This description is independent of the choices of the $\varpi_i$.*

(b) *For $i = 1, \dots, n$, let $\varphi_i$ denote the automorphism of $X$ induced by the absolute $p$-power Frobenius on $X_i$. For any fixed $i$, the quotient*

$$X/\Phi := X/\langle \varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n \rangle$$

*exists in* **Pfd** *and is canonically independent of the choice of $i$.*

(c) *If $X_1, \dots, X_n$ are qcqs, then $X/\Phi$ is qcqs.*

*Proof.* For these all assertions, it suffices to treat the case

$$X_i = \mathrm{Spa}(A_i, A_i^+)$$

with each $X_i$ affinoid perfectoid. Choose a topologically nilpotent unit $\varpi_i \in A_i$ for each $i$. Let

$$A^+ := \widehat{A_1^+ \otimes_{\mathbb{F}_p} \cdots \otimes_{\mathbb{F}_p} A_n^+}$$

where the completion is taken for the $(\varpi_1, \dots, \varpi_n)$-adic topology, and put

$$A := A^+[\varpi_1^{-1}, \dots, \varpi_n^{-1}].$$

Set

$$X := \left\{ x \in \mathrm{Spa}(A, A^+) \,\middle|\, |\varpi_i(x)| < 1 \text{ for } i = 1, \dots, n \right\}.$$

(a) Let $Y$ be a perfectoid space together with morphisms

$$f_i : Y \longrightarrow X_i.$$

The assertion is local on $Y$, so we may assume

$$Y = \mathrm{Spa}(B, B^+)$$

is affinoid perfectoid. Each $f_i$ corresponds to a morphism of Huber pairs

$$f_i^{\#} : A_i \longrightarrow B$$

sending $A_i^+$ into $B^+$. Multiplication in $B^+$ therefore gives a morphism

$$A_1^+ \otimes_{\mathbb{F}_p} \cdots \otimes_{\mathbb{F}_p} A_n^+ \longrightarrow B^+, \qquad a_1 \otimes \cdots \otimes a_n \longmapsto f_1^{\#}(a_1) \cdots f_n^{\#}(a_n).$$

The elements $f_i^{\#}(\varpi_i)$ are topologically nilpotent units in $B$. Hence the image of the ideal $(\varpi_1, \dots, \varpi_n)$ is topologically nilpotent, so the preceding morphism is continuous for the $(\varpi_1, \dots, \varpi_n)$-adic topology and extends uniquely to

$$A^+ \longrightarrow B^+.$$

Since each $f_i^{\#}(\varpi_i)$ is invertible in $B$, this further extends uniquely to

$$A \longrightarrow B.$$

We obtain a morphism

$$f : Y \longrightarrow \mathrm{Spa}(A, A^+).$$

For every $y \in Y$,

$$|\varpi_i(f(y))| = |f_i^{\#}(\varpi_i)(y)| < 1,$$

so the image of $f$ lies in $X$. By construction, the composite $Y \to X \to X_i$ is $f_i$ for every $i$. Conversely, any morphism $Y \to X$ gives the morphisms to the $X_i$ by composition. These constructions are inverse to each other, giving

$$\mathrm{Hom}_{\mathbf{Pfd}}(Y, X) \cong \prod_{i=1}^{n} \mathrm{Hom}_{\mathbf{Pfd}}(Y, X_i).$$

Thus $X$ is the categorical product of the $X_i$ in $\mathbf{Pfd}$.

(b) Fix $i = 1$ for the notation. For $x \in X$ and $j = 2, \dots, n$, put

$$r_j(x) := \frac{\log |\varpi_j(x)|}{\log |\varpi_1(x)|} \in \mathbb{R}_{>0}.$$

Choose rational numbers $a_j, b_j > 0$ such that

$$a_j \le r_j(x) \le b_j < p a_j.$$

Then $x$ belongs to the rational neighborhood

$$U = \left\{ y \in X \,\middle|\, |\varpi_1(y)|^{b_j} \le |\varpi_j(y)| \le |\varpi_1(y)|^{a_j} \text{ for } j = 2, \dots, n \right\}.$$

Under $\varphi_j$, the quantity $r_j$ is multiplied by $p$. Hence the translates of $U$ by

$$\Phi = \langle \varphi_2, \dots, \varphi_n \rangle$$

are pairwise disjoint, because

$$b_j < p a_j.$$

Thus the action of $\Phi$ on $X$ is totally discontinuous. The quotients of these neighborhoods by their stabilizers glue to a perfectoid space $X/\Phi$, proving (b).

(c) If the $X_i$ are qcqs, one may choose finitely many such neighborhoods whose $\Phi$-translates cover $X$. Their pairwise intersections are again of the same form. It follows that $X/\Phi$ is quasicompact and separated, hence qcqs.

□

In particular, Proposition 3.4 give the following identification:

*Remark.* Let $E = \mathbb{F}_q((\pi))$, let $F$ be an algebraically closed nonarchimedean field of characteristic $p$ containing $\mathbb{F}_q$, and let $L$ be a complete algebraically closed overfield of $E$. Then

$$\mathcal{X}_{F,E} \times_E L \cong \left(\mathrm{Spa}(F, \mathcal{O}_F) \times \mathrm{Spa}(L, \mathcal{O}_L)\right)/\Phi,$$

where the product is the categorical product in **Pfd** and $\Phi$ is the corresponding partial Frobenius action. Consequently, the universal geometric simple connectivity theorem implies that

$$\pi_1^{\mathrm{prof}}\left(\left(\mathrm{Spa}(F, \mathcal{O}_F) \times \mathrm{Spa}(L, \mathcal{O}_L)\right)/\Phi\right) = 1.$$

5.3. **The base change lemma.** Fix an algebraically closed nonarchimedean field $F$ of characteristic $p$. For $X \in \mathbf{Pfd}$, put

$$X_F := X \times \mathrm{Spa}(F, \mathcal{O}_F),$$

where the product is taken in **Pfd**, and let

$$\varphi_F : X_F \longrightarrow X_F$$

be the automorphism induced by the absolute $p$-power Frobenius on the second factor.

**Lemma 5.3** (Kedlaya's base change lemma)**.** *The base extension functor*

$$\mathbf{F\acute{E}t}(X) \longrightarrow \mathbf{F\acute{E}t}(X_F/\varphi_F)$$

*is an equivalence of categories.*

*Proof.* This is [8, Lemma 9.5]. Both

$$X \longmapsto \mathbf{F\acute{E}t}(X) \qquad \text{and} \qquad X \longmapsto \mathbf{F\acute{E}t}(X_F/\varphi_F)$$

are stacks for the étale topology. It therefore suffices to verify the assertion on geometric points.

Let

$$X = \mathrm{Spa}(L, L^+)$$

be a geometric point, with $L$ algebraically closed of characteristic $p$. Choose a topologically nilpotent unit $t \in L$ and put

$$E := \mathbb{F}_p((t)).$$

Then $L$ is a complete algebraically closed overfield of $E$. By the preceding remark,

$$X_F/\varphi_F \cong \mathcal{X}_{F,E} \times_E L.$$

The universal geometric simple connectivity theorem therefore implies that every finite étale cover of $X_F/\varphi_F$ splits completely. Since $L$ is algebraically closed, every finite étale cover of $X$ also splits completely. Thus both categories are canonically equivalent to the category of finite sets, and

$$\mathbf{F\acute{E}t}(X) \xrightarrow{\sim} \mathbf{F\acute{E}t}(X_F/\varphi_F).$$

Étale descent then gives the equivalence for arbitrary $X \in \mathbf{Pfd}$. □

## 6. Completion and infinite Galois descent

Let $K$ be a complete nonarchimedean field of rank one, let $K_\infty/K$ be a Galois algebraic extension, and put

$$\Gamma := \mathrm{Gal}(K_\infty/K), \qquad L := \widehat{K_\infty}.$$

Since $K$ is complete, its valuation is henselian and hence extends uniquely to $K_\infty$. In particular, every element of $\Gamma$ acts isometrically on $K_\infty$ and therefore extends uniquely to a continuous automorphism of $L$.

If a topological field $M$ is equipped with a continuous action of $\Gamma$, we write $\mathbf{F\acute{E}t}_\Gamma(M)$ for the category of finite étale $M$-algebras equipped with a compatible continuous semilinear $\Gamma$-action. Equivalently, we may regard this as the category of $\Gamma$-equivariant finite étale covers of $\mathrm{Spa}(M, \mathcal{O}_M)$.

6.1. **Completion.** We first record that passing from $K_\infty$ to its completion does not change its finite étale theory.

**Lemma 6.1.** *The completion functor induces an equivalence*

$$\mathbf{F\acute{E}t}(K_\infty) \xrightarrow{\sim} \mathbf{F\acute{E}t}(L).$$

*Moreover, this equivalence is $\Gamma$-equivariant and hence induces an equivalence*

$$\mathbf{F\acute{E}t}_\Gamma(K_\infty) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(L).$$

*Proof.* The field $K_\infty$ is henselian, since it is an algebraic extension of the henselian valued field $K$. In the terminology of Berkovich, it is therefore quasicomplete. By [11, Propositions 2.4.1 and 2.4.3], completion induces an equivalence between the categories of finite separable extensions of $K_\infty$ and of $L$.

More explicitly, if $M/K_\infty$ is finite separable, then its valuation extends uniquely from $K_\infty$, and

$$\widehat{M} \cong M \otimes_{K_\infty} L, \qquad [\widehat{M} : L] = [M : K_\infty].$$

Conversely, every finite separable extension of $L$ is obtained in this way from a finite separable extension of $K_\infty$; the essential surjectivity follows from Krasner's lemma. Taking finite products gives the asserted equivalence for finite étale algebras.

The construction is functorial with respect to isometric automorphisms of $K_\infty$. Thus for $\gamma \in \Gamma$, completion commutes with the pullback by $\gamma$, and the equivalence above is $\Gamma$-equivariant. A $\Gamma$-equivariant structure and its cocycle condition are therefore transported uniquely across the equivalence, giving

$$\mathbf{F\acute{E}t}_\Gamma(K_\infty) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(L).$$

□

6.2. **Infinite Galois descent.** We next recall the Galois descent statement that will be used in the proof of the main theorem.

**Lemma 6.2.** *Base extension induces an equivalence*

$$\mathbf{F\acute{E}t}(K) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(K_\infty).$$

*Proof.* The morphism

$$\mathrm{Spec}(K_\infty) \longrightarrow \mathrm{Spec}(K)$$

is faithfully flat and quasi-compact. Finite étale morphisms satisfy effective fpqc descent; see [13, Exposé IX, §§4–5]. Hence finite étale $K$-algebras are equivalent to finite étale $K_\infty$-algebras equipped with descent data relative to $K_\infty/K$.

Since $K_\infty/K$ is Galois, such descent data may equivalently be expressed as continuous semilinear $\Gamma$-actions. We briefly recall why the fact that $\Gamma$ may be infinite causes no difficulty. Write

$$K_\infty = \bigcup_M M,$$

where $M/K$ runs through the finite Galois subextensions. A finite étale $K_\infty$-algebra is of finite presentation, so it descends to some finite stage $M$. Likewise, all morphisms occurring in a descent datum descend after enlarging $M$. At a finite Galois stage, the usual identification

$$M \otimes_K M \cong \prod_{\sigma \in \mathrm{Gal}(M/K)} M$$

identifies a descent datum with a semilinear $\mathrm{Gal}(M/K)$-action satisfying the cocycle condition. Passing to the filtered union identifies descent data over $K_\infty/K$ with continuous semilinear $\Gamma$-actions. Thus

$$\mathbf{F\acute{E}t}(K) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(K_\infty).$$

□

## 7. Proof of the main theorem

7.1. **The construction and the main theorem.** Let $E$ be a complete rank-one nonarchimedean field of mixed characteristic $(0,p)$. Fix an algebraic closure $\overline{E}$ and put

$$C := \widehat{\overline{E}}.$$

Let $E_\infty/E$ be a Galois algebraic extension contained in $\overline{E}$, with profinite Galois group

$$\Gamma := \mathrm{Gal}(E_\infty/E).$$

Suppose that

$$L := \widehat{E_\infty}$$

is perfectoid. The action of $\Gamma$ on $E_\infty$ extends uniquely to a continuous faithful action on $L$, and hence induces a continuous faithful action on its tilt $L^\flat$.

Consider the categorical product in **Pfd**

$$P^\flat := \mathrm{Spa}(C^\flat, \mathcal{O}_{C^\flat}) \times \mathrm{Spa}(L^\flat, \mathcal{O}_{L^\flat}).$$

Let $\varphi_{C^\flat}$ and $\varphi_{L^\flat}$ denote the partial absolute Frobenii on the first and second factors, respectively, and put

$$V_L^\flat := P^\flat/\varphi_{L^\flat}^{\mathbb{Z}}, \qquad V_C^\flat := P^\flat/\varphi_{C^\flat}^{\mathbb{Z}}.$$

By Lemma 5.2(b), there is a canonical isomorphism

$$\iota : V_L^\flat \xrightarrow{\sim} V_C^\flat,$$

functorial in both factors. The group $\Gamma$ acts on the second factor of $P^\flat$ and commutes with both partial Frobenii; in particular, the preceding isomorphism is $\Gamma$-equivariant.

The structural morphism

$$V_L^\flat \longrightarrow \mathrm{Spa}(C^\flat, \mathcal{O}_{C^\flat})$$

allows us to untilt $V_L^\flat$ over $C$. Define

$$V := \left(V_L^\flat\right)^\sharp_C.$$

The action of $\Gamma$ on $V_L^\flat$ untilts to a continuous $C$-linear action on $V$. Finally, define

$$Z := Z(E_\infty/E; C) := V/\Gamma$$

as the sheaf quotient on the pro-étale site of perfectoid spaces over $C$.

**Theorem 7.1.** *There is a natural equivalence of Galois categories*

$$\mathbf{F\acute{E}t}(Z) \xrightarrow{\sim} \mathbf{F\acute{E}t}(E).$$

*In particular, after choosing compatible geometric base points,*

$$\pi_1^{\text{ét}}(Z) \cong G_E.$$

*Proof.* The desired equivalence is the composite

$$\mathbf{F\acute{E}t}(Z) \overset{(1)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(V) \overset{(2)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(V_L^\flat) \overset{(3)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(L^\flat)$$
$$\overset{(4)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(L) \overset{(5)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(E_\infty) \overset{(6)}{\simeq} \mathbf{F\acute{E}t}(E).$$

For (1), we first show that the action of $\Gamma$ on $V$ is free on geometric points. Let $C'/C$ be a complete algebraically closed extension and suppose that $\gamma \in \Gamma$ fixes a point

$$\overline{v} \in V(C').$$

By the tilting equivalence and the canonical isomorphism $\iota$, we may regard $\overline{v}$ as a point of $V_C^\flat(C'^\flat)$. After replacing $C'$ by a further complete algebraically closed extension if necessary, choose a lift

$$\overline{p} \in P^\flat(C'^\flat)$$

of $\overline{v}$. Since

$$V_C^\flat = P^\flat/\varphi_{C^\flat}^{\mathbb{Z}},$$

the equality $\gamma\overline{v} = \overline{v}$ implies

$$\gamma\overline{p} = \varphi_{C^\flat}^m \overline{p}$$

for some $m \in \mathbb{Z}$. Projecting to the second factor gives a continuous embedding

$$j : L^\flat \hookrightarrow C'^\flat$$

such that

$$j \circ \gamma = j,$$

because $\varphi_{C^\flat}$ acts only on the first factor. Since $j$ is injective, $\gamma$ acts trivially on $L^\flat$. The action of $\Gamma$ on $L^\flat$ is faithful, hence $\gamma = 1$. Thus the action of $\Gamma$ on $V(C')$ is free for every complete algebraically closed extension $C'/C$.

By [9, Proposition 4.3.2], the quotient map

$$V \longrightarrow Z = V/\Gamma$$

is therefore a pro-étale $\Gamma$-torsor. Effective descent for finite étale covers along such torsors [9, Proposition 4.3.3] gives

$$\mathbf{F\acute{E}t}(Z) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(V).$$

The equivalence (2) is the tilting equivalence

$$\mathbf{F\acute{E}t}(V) \simeq \mathbf{F\acute{E}t}(V_L^\flat).$$

Its functoriality with respect to automorphisms identifies the corresponding $\Gamma$-equivariant structures.

For (3), use first the canonical $\Gamma$-equivariant isomorphism

$$V_L^\flat \xrightarrow{\sim} V_C^\flat.$$

Apply Lemma 5.3, with

$$X = \mathrm{Spa}(L^\flat, \mathcal{O}_{L^\flat})$$

and with the algebraically closed perfectoid field $C^\flat$. The product occurring there is precisely $P^\flat$, and quotienting by the partial Frobenius on the $C^\flat$-factor gives $V_C^\flat$. Hence

$$\mathbf{F\acute{E}t}(L^\flat) \xrightarrow{\sim} \mathbf{F\acute{E}t}(V_C^\flat) \xrightarrow{\sim} \mathbf{F\acute{E}t}(V_L^\flat).$$

These equivalences are functorial in automorphisms of $L^\flat$, and therefore induce

$$\mathbf{F\acute{E}t}_\Gamma(V_L^\flat) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(L^\flat).$$

The equivalence (4) follows from the tilting equivalence for the perfectoid field $L$ and its tilt $L^\flat$. Again, functoriality identifies the $\Gamma$-actions.

Finally, the equivalence (5) follows from Lemma 6.1 and (6) follows from Lemma 6.2. □

7.2. **The case of $\mathbb{Q}_p\{\{X\}\}$.** We now apply Theorem 7.1 to

$$E = \mathbb{Q}_p\{\{X\}\}.$$

Choose compatible systems of $p$-power roots of unity and $p$-power roots of $X$, and put

$$E_n := E(\mu_{p^n}, X^{1/p^n}), \qquad E_\infty := \bigcup_{n\geq 1} E_n, \qquad \Gamma_E := \mathrm{Gal}(E_\infty/E).$$

This is the one-dimensional $p$-basis case of the Kummer tower considered by Scholl [14, §2.3]. It is strictly deeply ramified; consequently, its completion

$$L := \widehat{E_\infty}$$

is a perfectoid field by [10, Proposition 6.6.6]; see also [12, Remark 3.3].

Put $C := \widehat{\overline{E}}$, and form

$$P^\flat = \mathrm{Spa}(C^\flat, \mathcal{O}_{C^\flat}) \times \mathrm{Spa}(L^\flat, \mathcal{O}_{L^\flat}), \qquad V_L^\flat = P^\flat/\varphi_{L^\flat}^{\mathbb{Z}}.$$

Let

$$V := \big(V_L^\flat\big)_C^\sharp, \qquad Z_E := V/\Gamma_E.$$

Then Theorem 7.1 gives

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}\big(\mathbb{Q}_p\{\{X\}\}\big),$$

and hence

$$\pi_1^{\text{ét}}(Z_E) \cong G_{\mathbb{Q}_p\{\{X\}\}}.$$

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